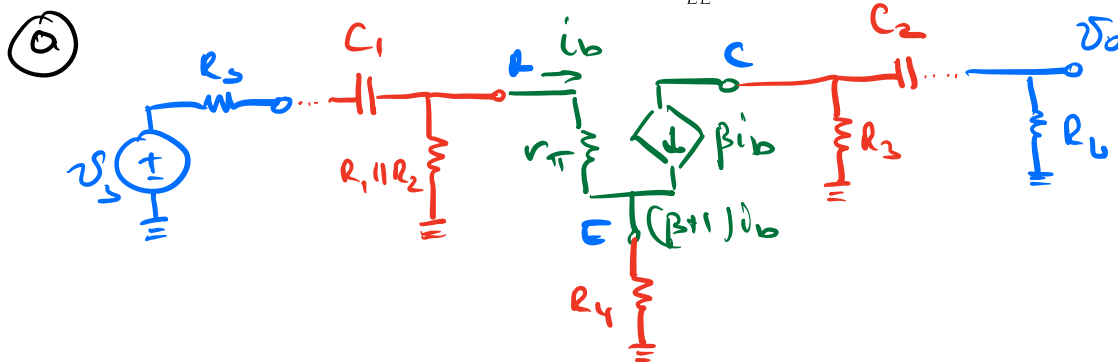
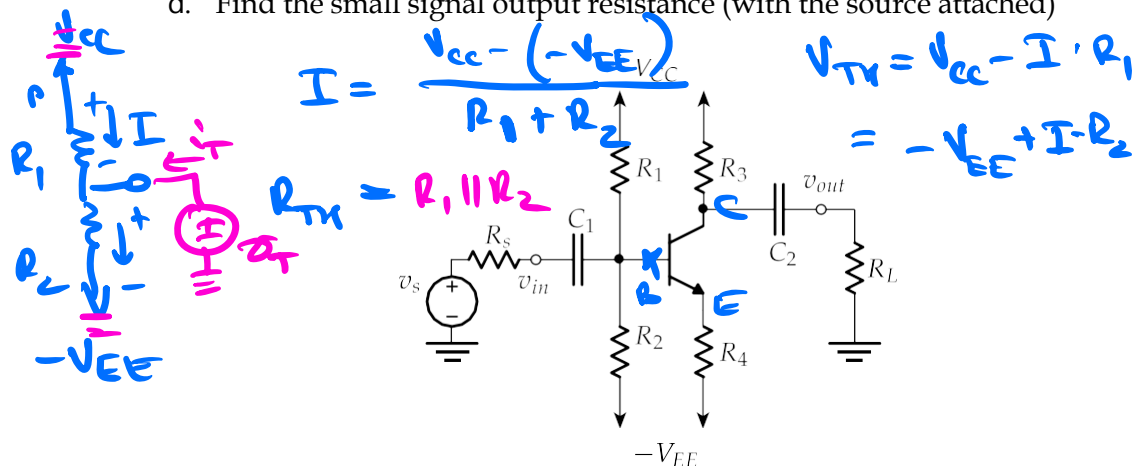


Topic: BJT Small Signal Analysis

$$\frac{R_1}{R_1 + R_2}$$

1. (15 points): For the following circuit,

- Draw the small signal model of the circuit
- Find the amplifier's ac gain (v_{out}/v_{in} with source and load removed)
- Find the small signal input resistance (with the load attached)
- Find the small signal output resistance (with the source attached)



source and load disconnected

(b)

$$\frac{v_o}{v_{in}} = \frac{-\beta i_b R_3}{r_{\pi} i_b + (\beta + 1) i_b R_4} = -\frac{\beta R_3}{r_{\pi} + (\beta + 1) R_4}$$

(c) connect load

$$R_{in} = R_1 || R_2 || \frac{r_{\pi} i_b + (\beta + 1) i_b R_4}{i_b} = R_1 || R_2 || (r_{\pi} + (\beta + 1) R_4)$$

(d) connect input, short v_s

$$i_b \cdot R_3 || R_1 || R_2 + i_b r_{\pi} + (\beta + 1) i_b R_4 = 0 \Rightarrow i_b = 0$$

$$R_o = R_3$$